Lab 2: Exploratory Data Analysis & Distribution Fitting

Department of Electrical Engineering, IIT Madras

Demo date: Feb 11, 2026 in class

Session Goal

Perform **Exploratory Data Analysis (EDA)** on real medical data and apply **estimation techniques** (Method of Moments vs Maximum Likelihood) to model skewed datasets.

1 Preparation & Setup

- **Submission:** One Colab notebook (`.ipynb`) per group. You can add to the same notebook you had for Lab 1.
- **Libraries:** `pandas`, `numpy`, `matplotlib.pyplot`, `seaborn`, `scipy.stats`.

2 Data Collection

The data used is from the Berkeley Stat Labs repository (sincere thanks to their efforts) at `https://www.stat.berkeley.edu/users/statlabs/labs.html`.

3 Part A: Maternal Smoking (EDA)

Context: The Child Health and Development Studies (CHDS) investigated the effects of maternal smoking on pregnancy outcomes. This is listed under "Maternal Smoking and Infant Health II".

Tasks

1. **Data Cleaning:** Load the data to a pandas dataframe. Clean up data that is not available.
2. **Data description:** Describe the data as a short table. What are the columns?
3. **Visual Comparison:**
   - Create a single figure containing two **Histograms** (overlaid with suitable transparency) comparing birth weights.
   - Create a **Boxplot** with `smoke` on the x-axis and `bwt` on the y-axis.
4. **Summary Statistics:** Calculate and print the Mean, Median, Standard Deviation, and Skewness (learn about skewness and Kurtosis) of birth weights for both groups.
5. **Inference:** Based *only* on the graphs and stats (no hypothesis tests yet), does it look like smoking affects birth weight? Write your conclusion in a text cell.

6. **Explore:** Explore the data in your own way and make other such plots to compare different groups.

---

4 Part B: Video Game Analytics (Distribution Fitting)

Context: A survey of time spent playing video games in the week prior to an exam. This is listed under "Video game survey".

- **Dataset:** `video.data`
- **Key Variable:** `time` (Hours played).

Tasks

1. **The "Zero" Problem:** Many students played 0 hours, creating a spike at zero. For this exercise, **filter out the zeros** and only analyze students who played at least some video games (`time > 0`).
2. **Method of Moments (MOM):**
   - Assume the data follows a **Gamma Distribution**.
   - Derive MOM estimates for shape (α) and scale (β).
3. **Maximum Likelihood Estimation (MLE):**
   - Estimate parameters α and β using your own MLE. Compare with scipy.stats.gamma.fit.
4. **Visual Fit:**
   - Plot a density **Histogram** of the non-zero data.
   - Overlay the **Gamma PDF** using your MOM parameters.
   - Overlay the **Gamma PDF** using the MLE parameters.
   - *Question:* Which method fits the "peak" better visually?

---

5 The Demo Session (Assessment)

- Ensure your code is clean and reproducible.
- **Live Alteration:** The TAs will provide a specific modification to your analysis (e.g., analyzing a different variable or fitting a different distribution). You will need to modify your code live and interpret the new results.